

THE CUBIC THRESHOLD FOR COEFFICIENTWISE LOG-CONCAVITY OF MULTIPLE-POCHHAMMER SERIES

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ABSTRACT. Multiple-Pochhammer power series associated with a nonnegative log-concave sequence with no internal zeros were shown by Karp and Zhang to be coefficientwise log-concave at multiplicity two, while the cubic case remained open. We prove the cubic conjecture, in the equivalent fixed-sum form of coefficientwise Schur-concavity. Together with a degree-three obstruction for larger multiplicities, this yields the sharp classification: universal coefficientwise log-concavity holds precisely at multiplicities two and three. The proof reduces the coefficient inequalities to monotonicity of a symmetric residue kernel and shows that, in the cubic case, this kernel preserves the central ordering imposed by log-concavity.

1. INTRODUCTION

For an integer $r \geq 2$, consider the modified multiple-Pochhammer series

$$(1.1) \quad F_{f,r}(\mu; x) = \sum_{n \geq 1} f_n \frac{(\mu)_{rn}}{(rn-1)!} x^n,$$

where (f_n) is a nonnegative log-concave sequence with no internal zeros. Karp and Zhang proved coefficientwise log-concavity for $r = 2$ [6, Thm. 2], gave numerical counterexamples for $r \geq 4$ [6, Rem. 5], and formulated the cubic case $r = 3$ as their Conjecture 1. Karp and Zhang report that A. Ninh suggested the construction to them and conjectured log-concavity for the resulting family [6, §3]; related results on log-concavity and Turán-type inequalities appear in [2, 3, 5, 9].

A parameter-dependent formal power series $F(\mu; x)$ is *coefficientwise log-concave* if, for every $\mu, \alpha, \beta \geq 0$, all coefficients of

$$F(\mu + \alpha; x)F(\mu + \beta; x) - F(\mu; x)F(\mu + \alpha + \beta; x)$$

are nonnegative. This is stronger than pointwise log-concavity wherever the series converges. Here “log-concave with no internal zeros” means that $f_n^2 \geq f_{n-1}f_{n+1}$ wherever defined and that the positive support is an interval of integers. These are Karp and Zhang’s hypotheses [6, Def. 1]; the interval-support condition is essential (Remark 2.2). Except in Corollary 6.1, the arguments are formal and require no growth assumption. We use $(\mu)_j$ for the rising factorial.

1.1. Main results. We prove the missing cubic case in an equivalent, base-point-free form. A symmetric function of two variables is Schur-concave [8, Def. 3.A.1] exactly when, at each fixed sum, it decreases as the two variables move apart.

Theorem 1.1 (Cubic coefficientwise Schur-concavity). *Let $(f_n)_{n \geq 1}$ be a nonnegative log-concave sequence with no internal zeros, and put*

$$(1.2) \quad F_f(\mu; x) = \sum_{n \geq 1} f_n \frac{(\mu)_{3n}}{(3n-1)!} x^n.$$

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For every $s > 0$ and every $0 \leq d_1 \leq d_2 \leq s/2$, the formal power series

$$(1.3) \quad \begin{aligned} \mathcal{S}_f(s; d_1, d_2; x) = & F_f\left(\frac{s}{2} + d_1; x\right) F_f\left(\frac{s}{2} - d_1; x\right) \\ & - F_f\left(\frac{s}{2} + d_2; x\right) F_f\left(\frac{s}{2} - d_2; x\right) \end{aligned}$$

has nonnegative coefficients. Equivalently, for every $\mu, \alpha, \beta \geq 0$, the generalized Turánian

$$(1.4) \quad \Delta_f(\mu; \alpha, \beta; x) = F_f(\mu + \alpha; x)F_f(\mu + \beta; x) - F_f(\mu; x)F_f(\mu + \alpha + \beta; x)$$

has nonnegative coefficients.

The fixed-sum statement is equivalent to the generalized Turánian formulation. A triple (μ, α, β) corresponds to

$$s = 2\mu + \alpha + \beta, \quad d_1 = \frac{|\alpha - \beta|}{2}, \quad d_2 = \frac{\alpha + \beta}{2},$$

and conversely

$$\mu = \frac{s}{2} - d_2, \quad \alpha = d_2 + d_1, \quad \beta = d_2 - d_1.$$

The inverse correspondence is determined up to the symmetry $\alpha \leftrightarrow \beta$. Thus the two families of coefficient inequalities are identical, including the boundary case $\mu = 0$.

Together with the quadratic theorem of Karp and Zhang, the cubic result yields the sharp multiplicity classification: universal coefficientwise log-concavity holds exactly for $r = 2, 3$ (Corollary 5.2). For every $r \geq 4$, failure already occurs in degree three for a two-term log-concave sequence (Proposition 5.1).

1.2. Central-slope cancellation at multiplicity three. The threshold is not accidental. At degree three, the derivative at the center of the constant-weight residue kernel contains the factor $r(3 - r)$; see (5.1). The quadratic case has ordinary central curvature; for every $r \geq 4$, the curvature reverses sign and gives an obstruction. At $r = 3$ the quadratic term vanishes exactly: the center is a quartically flat maximum. The main work of the paper is to show that this critical local cancellation extends to global kernel monotonicity in every degree and for every admissible weight sequence.

1.3. Proof strategy and organization. The proof aligns one notion of centrality at three levels. On the sequence side, log-concavity with interval support is equivalent to central increase of every anti-diagonal product sequence $f_k f_{m-k}$ (Lemma 2.1). On the parameter side, a beta-binomial representation turns the degree- m coefficient into an expectation of a symmetric residue kernel in the coordinate $q = p(1-p)$; increasing parameter imbalance moves this coordinate away from the center in likelihood-ratio order (Lemma 2.3). The cubic-specific input is therefore the kernel theorem Theorem 3.1: centrally increasing symmetric weights produce a kernel increasing toward $p = 1/2$. Beta concentration gives the converse implication, so the kernel condition is an exact reduction rather than only a sufficient device (Proposition 2.4).

Section 2 establishes this exact reduction. Section 3 proves the cubic kernel theorem, first for constant weights and then for the full monotone-weight cone. Section 4 assembles the proof of Theorem 1.1 and determines its strict coefficient support. Section 5 proves the sharp multiplicity threshold, and Section 6 records analytic and hypergeometric consequences. Finally, Section 7 summarizes what remains beyond the universal threshold.

2. EXACT REDUCTION TO A RESIDUE KERNEL

The sequence condition and the parameter comparison can both be converted into centrality statements. On the sequence side, Levit and Mandrescu [7, Lem. 2.2(ii)] give a cross-product characterization of nonnegative log-concave functions without internal zeros; the following adjacent anti-diagonal form is the interface needed here.

Lemma 2.1. *Let $(f_n)_{n \geq 1}$ be a nonzero nonnegative sequence. The following are equivalent.*

- (1) (f_n) is log-concave and has no internal zeros.

(2) For every $m \geq 2$, the anti-diagonal products are nondecreasing toward the center:

$$(2.1) \quad f_k f_{m-k} \leq f_{k+1} f_{m-k-1} \quad (1 \leq k < \lfloor m/2 \rfloor).$$

Proof. Assume first that f is log-concave with interval support. If $f_k f_{m-k} = 0$, (2.1) is immediate. Otherwise interval support gives $f_j > 0$ for $k \leq j \leq m-k$. With $\rho_j = f_{j+1}/f_j$, log-concavity makes (ρ_j) nonincreasing, and $k \leq m-k-1$ gives

$$\frac{f_{k+1} f_{m-k-1}}{f_k f_{m-k}} = \frac{\rho_k}{\rho_{m-k-1}} \geq 1.$$

Conversely, assume (2.1) for every m . Taking $m = 2j$ and $k = j-1$ gives $f_j^2 \geq f_{j-1} f_{j+1}$, so f is log-concave. If $f_a, f_b > 0$ with $b \geq a+2$, take $m = a+b$ and $k = a$. Then

$$f_{a+1} f_{b-1} \geq f_a f_b > 0.$$

Thus both inward neighbors are positive; iteration fills every gap between two positive terms. Hence the positive support is an interval. $\square$

Remark 2.2. The central-product characterization also explains why interval support is essential. A gap in the positive support forces some anti-diagonal product sequence outside the centrally nondecreasing cone: if $f_a, f_b > 0$ but an intermediate term vanishes, the inward inequality used in the converse of Lemma 2.1 must fail somewhere. This is not merely a limitation of the proof. Take $f_1 = f_4 = 1$ and $f_n = 0$ otherwise. The log-concavity inequalities still hold, but the support $\{1, 4\}$ is not an interval. At $m = 5$ the anti-diagonal products are $f_1 f_4 = f_4 f_1 = 1$ and $f_2 f_3 = f_3 f_2 = 0$, so they decrease toward the center, and

$$[x^5] \Delta_f(2; 1, 1; x) = -9360.$$

Thus failure of sequence centrality can produce an actual sign reversal.

We now pass from anti-diagonal weights to the parameter comparison, keeping the weights arbitrary for the moment. Fix $m \geq 2$ and nonnegative weights $(w_k)_{1 \leq k \leq m-1}$. For $u, v \geq 0$, define

$$(2.2) \quad C_{m,w}(u, v) = \sum_{k=1}^{m-1} w_k \frac{(u)_{3k} (v)_{3(m-k)}}{(3k-1)! [3(m-k)-1]!}.$$

Both $C_{m,w}(0, v)$ and $C_{m,w}(u, 0)$ vanish. For $u, v > 0$, put $N = 3m$ and let J have the beta-binomial law

$$\mathbb{P}(J = j) = \binom{N}{j} \frac{(u)_j (v)_{N-j}}{(u+v)_N}, \quad 0 \leq j \leq N.$$

Since

$$\frac{(u)_j (v)_{N-j}}{(j-1)!(N-j-1)!} = \frac{(u+v)_N}{N!} j(N-j) \mathbb{P}(J = j),$$

defining

$$\tilde{w}_j = \begin{cases} w_{j/3}, & 0 < j < N \text{ and } 3 \mid j, \\ 0, & \text{otherwise,} \end{cases}$$

gives

$$(2.3) \quad C_{m,w}(u, v) = \frac{(u+v)_N}{N!} \mathbb{E}[J(N-J) \tilde{w}_J].$$

The endpoint terms vanish because of the factor $J(N-J)$.

Using the standard mixture $P \sim \text{Beta}(u, v)$, $J \mid P \sim \text{Bin}(N, P)$, together with

$$j(N-j) \binom{N}{j} = N(N-1) \binom{N-2}{j-1},$$

we obtain

$$(2.4) \quad C_{m,w}(u, v) = \frac{(u+v)_N}{N!} N(N-1) \mathbb{E} G_{m,w}(P),$$

where

$$(2.5) \quad G_{m,w}(p) = \sum_{k=1}^{m-1} w_k \binom{3m-2}{3k-1} p^{3k} (1-p)^{3(m-k)}, \quad 0 \leq p \leq 1.$$

The shift from $3k$ to $3k-1$ comes from $j-1 \equiv 2 \pmod{3}$. If $w_k = w_{m-k}$, then $G_{m,w}(p) = G_{m,w}(1-p)$. In that case write $\hat{G}_{m,w}(q)$ for the common value of $G_{m,w}(p)$ at the two points satisfying $p(1-p) = q$, $0 \leq q \leq 1/4$. For fixed $s > 0$, let

$$P_d \sim \text{Beta}\left(\frac{s}{2} + d, \frac{s}{2} - d\right), \quad Q_d = P_d(1 - P_d).$$

Then (2.4) becomes

$$(2.6) \quad C_{m,w}\left(\frac{s}{2} + d, \frac{s}{2} - d\right) = \kappa_{m,s} \mathbb{E} \hat{G}_{m,w}(Q_d), \quad \kappa_{m,s} = \frac{(s)_{3m}}{(3m)!} 3m(3m-1) > 0.$$

Thus $q = p(1-p)$ is the common centrality coordinate for the residue kernel and the beta mixture: kernel monotonicity toward $p = 1/2$ is exactly monotonicity of $\hat{G}_{m,w}$ in q , while Lemma 2.3 will show that parameter imbalance moves Q_d in the opposite direction.

At fixed parameter sum, imbalance is ordered monotonically by the beta mixture. The comparison is a likelihood-ratio order; see [11, §1.C]. It compares the symmetric transforms $P(1-P)$; since $p \mapsto p(1-p)$ increases on $[0, 1/2]$, this is the folded-beta comparison of [1, Thm. 3], whose hypotheses reduce at a fixed parameter sum to $2d_1, 2d_2 \in \{0\} \cup [1, \infty)$. The direct proof below removes that restriction, giving the comparison for every $0 \leq d_1 < d_2 < s/2$.

Lemma 2.3. *Fix $s > 0$. For $0 \leq d < s/2$, let $P_d \sim \text{Beta}(s/2 + d, s/2 - d)$ and $Q_d = P_d(1 - P_d)$. If $0 \leq d_1 < d_2 < s/2$, then $Q_{d_2} \leq_{\text{lr}} Q_{d_1}$, where $X \leq_{\text{lr}} Y$ means that the ratio of the density of Y to the density of X is nondecreasing on their common support. Consequently, if $H : [0, 1] \rightarrow \mathbb{R}$ is symmetric, $H(p) = H(1-p)$, and nondecreasing on $[0, 1/2]$, then*

$$(2.7) \quad d_1 \leq d_2 \implies \mathbb{E}H(P_{d_1}) \geq \mathbb{E}H(P_{d_2}).$$

If H is strictly increasing on $(0, 1/2)$ and $d_1 < d_2$, then the inequality is strict.

Proof. Such an H is bounded and Borel measurable, so the expectations in (2.7) exist. The case $d_1 = d_2$ is immediate, so assume $d_1 < d_2$. Here $Q_d \in (0, 1/4)$. For

$$p_{\pm}(q) = \frac{1 \pm \sqrt{1-4q}}{2}, \quad \ell(q) = \log \frac{p_+(q)}{p_-(q)},$$

the two preimages of q satisfy $p_+p_- = q$ and $p_- = 1 - p_+$, so they contribute $p_+^{s/2+d-1}p_-^{s/2-d-1} + p_-^{s/2+d-1}p_+^{s/2-d-1} = 2q^{s/2-1} \cosh(d\ell(q))$. Since $|dq/dp| = \sqrt{1-4q}$, the density of Q_d is

$$g_d(q) = \frac{2}{B(s/2 + d, s/2 - d)} q^{s/2-1} (1-4q)^{-1/2} \cosh(d\ell(q)).$$

The densities g_{d_1} and g_{d_2} share the factor $q^{s/2-1}(1-4q)^{-1/2}$, so

$$\frac{g_{d_2}(q)}{g_{d_1}(q)} = \frac{B(s/2 + d_1, s/2 - d_1)}{B(s/2 + d_2, s/2 - d_2)} \frac{\cosh(d_2\ell(q))}{\cosh(d_1\ell(q))}.$$

The function ℓ is strictly decreasing, since $p_+(q)/p_-(q)$ decreases to 1 as $q \uparrow 1/4$. For $d_2 > d_1$, $\cosh(d_2\ell)/\cosh(d_1\ell)$ is strictly increasing in $\ell > 0$, because $d \mapsto d \tanh(d\ell)$ is strictly increasing; hence g_{d_2}/g_{d_1} is strictly decreasing in q . Thus $Q_{d_2} \leq_{\text{lr}} Q_{d_1}$. Since both densities integrate to one, the strictly monotone ratio crosses 1 exactly once; hence $g_{d_1} - g_{d_2}$ changes sign once, from negative to positive, and for every nondecreasing ϕ ,

$$\mathbb{E}\phi(Q_{d_1}) - \mathbb{E}\phi(Q_{d_2}) = \int_0^{1/4} (\phi(q) - \phi(q^*)) (g_{d_1}(q) - g_{d_2}(q)) dq \geq 0,$$

where q^* is the crossing point and the integrand is nonnegative throughout. Finally, symmetry of H gives $H(p) = \tilde{H}(p(1-p))$, with $\tilde{H}$ nondecreasing on $[0, 1/4]$ precisely when H is nondecreasing on $[0, 1/2]$; taking $\phi = \tilde{H}$ proves (2.7). If H is strictly increasing and $d_1 < d_2$, then $\tilde{H}$ is strictly

increasing and $g_{d_1} - g_{d_2}$ vanishes only at q^* ; hence the integrand in the preceding display is positive off q^* , and the inequality is strict. $\square$

The beta representation and the likelihood-ratio order give one direction of the reduction. Concentration of beta laws gives the converse.

Proposition 2.4. *Fix $m \geq 2$ and symmetric nonnegative weights $w_k = w_{m-k}$. Then the following are equivalent:*

- (1) $G_{m,w}$ is nondecreasing on $[0, 1/2]$;
- (2) $(u, v) \mapsto C_{m,w}(u, v)$ is Schur-concave on $[0, \infty)^2$.

If $G_{m,w}$ is strictly increasing and the weights are not all zero, then at each fixed $s > 0$,

$$d \mapsto C_{m,w}\left(\frac{s}{2} + d, \frac{s}{2} - d\right)$$

is strictly decreasing on $[0, s/2]$.

Proof. Assume first that $G_{m,w}$ is nondecreasing on $[0, 1/2]$. For $0 \leq d < s/2$, the fixed-sum representation (2.6) and Lemma 2.3 give the required monotonicity; if $G_{m,w}$ is strictly increasing, the inequality is strict. At the endpoint, $C_{m,w}(s, 0) = 0$ by (2.2), while every interior value is nonnegative and is positive when the weights are not all zero. Symmetry completes the Schur-concavity statement.

Conversely, suppose $C_{m,w}$ is Schur-concave for every fixed sum. Choose $1/2 \leq p_1 < p_2 < 1$, and for $s > 0$ let

$$P_{s,i} \sim \text{Beta}(sp_i, s(1 - p_i)), \quad i = 1, 2.$$

The corresponding imbalances are $d_i = s(p_i - 1/2)$. Since the prefactor in (2.4) is common at fixed s , Schur-concavity gives

$$\mathbb{E}G_{m,w}(P_{s,1}) \geq \mathbb{E}G_{m,w}(P_{s,2}).$$

But

$$\mathbb{E}P_{s,i} = p_i, \quad \text{Var}(P_{s,i}) = \frac{p_i(1 - p_i)}{s + 1},$$

so $P_{s,i} \rightarrow p_i$ in probability as $s \rightarrow \infty$. Since $G_{m,w}$ is a bounded continuous polynomial on $[0, 1]$, letting $s \rightarrow \infty$ yields $G_{m,w}(p_1) \geq G_{m,w}(p_2)$. Thus $G_{m,w}$ is nonincreasing on $[1/2, 1]$, hence by continuity on $[1/2, 1]$. Symmetry makes it nondecreasing on $[0, 1/2]$. $\square$

Kernel monotonicity is therefore precisely the analytic obstruction to coefficientwise Schur-concavity at a fixed degree.

Remark 2.5. For the weights $w_k = f_k f_{m-k}$ arising from Theorem 1.1,

$$C_{m,w}(u, v) = [x^m] F_f(u; x) F_f(v; x).$$

Assume $f \not\equiv 0$ and that $F_f(u; x)$ and $F_f(v; x)$ converge at some $x > 0$, with $u, v > 0$. Let N_u, N_v be independent with

$$\mathbb{P}_x(N_u = n) = \frac{f_n(u) 3n x^n}{(3n - 1)! F_f(u; x)}, \quad n \geq 1,$$

and analogously for N_v , and write $I = \{n \geq 1 : f_n > 0\}$. If $m \notin I + I$, then $C_{m,w}(u, v) = 0$ and the degree- m sector vanishes. If $m \in I + I$, then $C_{m,w}(u, v) > 0$, the conditioning event has positive probability, and the power of x cancels under conditioning:

$$\mathbb{P}_x(N_u = k \mid N_u + N_v = m) = \frac{f_k f_{m-k}(u) 3k(v) 3(m-k)}{(3k - 1)! [3(m - k) - 1]! C_{m,w}(u, v)}.$$

Thus, degree by degree, Theorem 1.1 is a microcanonical Schur-concavity statement; summing these fixed-total inequalities against x^m yields the ordinary pointwise Turán inequality. The beta-binomial identity (2.3) makes these fixed-total sectors explicit, while the fixed-sum form (2.6) compares parameter imbalance by stochastic order.

3. THE CUBIC RESIDUE KERNEL

The exact reduction leaves one genuinely cubic question: whether the residue kernel preserves central order. For symmetric nonnegative weights $w_k = w_{m-k}$, the kernel of (2.5) is

$$G_{m,w}(p) = \sum_{k=1}^{m-1} w_k \binom{3m-2}{3k-1} p^{3k} (1-p)^{3(m-k)}, \quad G_{m,w}(p) = G_{m,w}(1-p).$$

Theorem 3.1 (Weighted cubic residue monotonicity). *Suppose*

$$(3.1) \quad 0 \leq w_1 \leq w_2 \leq \cdots \leq w_{\lfloor m/2 \rfloor}, \quad w_k = w_{m-k}.$$

Then $G_{m,w}$ is nondecreasing on $[0, 1/2]$. If the weights are not all zero, then $G_{m,w}$ is strictly increasing on $[0, 1/2]$.

3.1. The constant-weight kernel. Let $w_k \equiv 1$, and write $G_m = G_{m,1}$. Set $n = 3m - 2$ and $t = p/(1-p)$. For $0 \leq p \leq 1/2$ one has $0 \leq t \leq 1$, and

$$G_m\left(\frac{t}{1+t}\right) = \frac{1}{(1+t)^{3m}} \sum_{k=1}^{m-1} \binom{n}{3k-1} t^{3k}.$$

Differentiation gives

$$(3.2) \quad \frac{d}{dt} G_m\left(\frac{t}{1+t}\right) = \frac{3J_m(t)}{(1+t)^{3m+1}},$$

where

$$(3.3) \quad J_m(t) = \sum_{k=1}^{m-1} \binom{n}{3k-1} t^{3k-1} (k - (m-k)t).$$

Lemma 3.2 (Bernstein-basis positivity certificate). *For every $m \geq 2$, with $n = 3m - 2$, write*

$$J_m(t) = \sum_{j=0}^{n+1} b_j \binom{n+1}{j} t^j (1-t)^{n+1-j}$$

in the degree- $(n+1)$ Bernstein basis. Then $b_j \geq 0$ for every $0 \leq j \leq n+1$, and

$$b_2 = \frac{n-1}{n+1} > 0.$$

Consequently, $J_m(t) > 0$ for $0 < t < 1$, and G_m is strictly increasing on $[0, 1/2]$.

Proof. Writing $\nu = 3k - 1$ in (3.3) gives

$$J_m(t) = \frac{1}{3} \sum_{\substack{0 \leq \nu \leq n \\ \nu \equiv 2(3)}} \binom{n}{\nu} t^\nu ((\nu+1) - (n-\nu+1)t).$$

To certify positivity on $[0, 1]$, we compute the degree- $(n+1)$ Bernstein coefficients of J_m . The projective substitution $t = \xi/(1+\xi)$ is chosen because

$$(1+\xi)^{n+1} t^j (1-t)^{n+1-j} = \xi^j;$$

thus it converts the Bernstein basis into the ordinary monomial basis. Introduce

$$(3.4) \quad \mathcal{P}_m(\xi) = (1+\xi)^{n+1} J_m\left(\frac{\xi}{1+\xi}\right).$$

Then

$$(3.5) \quad \mathcal{P}_m(\xi) = \frac{1}{3} \sum_{\nu \equiv 2(3)} \binom{n}{\nu} \xi^\nu (1+\xi)^{n-\nu} ((\nu+1) + (2\nu-n)\xi).$$

For $0 \leq j \leq n+1$, coefficient extraction from (3.5) gives

$$\begin{aligned} [\xi^j] \mathcal{P}_m(\xi) &= \frac{1}{3} \sum_{\substack{0 \leq \nu \leq j \\ \nu \equiv 2 \pmod{3}}} \binom{n}{\nu} \left[(\nu+1) \binom{n-\nu}{j-\nu} + (2\nu-n) \binom{n-\nu}{j-\nu-1} \right] \\ (3.6) \quad &= \frac{\binom{n+1}{j}}{3(n+1)} S_{n,j}, \end{aligned}$$

where

$$(3.7) \quad S_{n,j} = \sum_{\substack{0 \leq \nu \leq j \\ \nu \equiv 2 \pmod{3}}} \binom{j}{\nu} (n-\nu+1)(2\nu+1-j).$$

The second equality in (3.6) follows from the elementary identities

$$\binom{n}{\nu} \binom{n-\nu}{j-\nu} = \binom{n}{j} \binom{j}{\nu}, \quad \binom{n}{\nu} \binom{n-\nu}{j-\nu-1} = \binom{n+1}{j} \binom{j}{\nu} \frac{j-\nu}{n+1},$$

the second of which is the first at $j-1$, after $\binom{n}{j-1} = \binom{n+1}{j} \frac{j}{n+1}$ and $\binom{j-1}{\nu} = \binom{j}{\nu} \frac{j-\nu}{j}$. For the coefficients b_j in the Bernstein expansion of Lemma 3.2, the defining property of the transform gives $[\xi^j] \mathcal{P}_m(\xi) = b_j \binom{n+1}{j}$; hence (3.6) yields

$$b_j = \frac{S_{n,j}}{3(n+1)}.$$

Since every Bernstein basis function is nonnegative on $[0, 1]$, nonnegative b_j directly certify $J_m(t) \geq 0$ there; see [10] for background on positivity representations on an interval. It remains to prove that every $S_{n,j}$ is nonnegative.

Put $\delta = n+1-j$. A third-root-of-unity filter evaluates the sum over the class $\nu \equiv 2 \pmod{3}$. With $\omega = e^{2\pi i/3}$, one has $1 + \omega = e^{i\pi/3}$, so the nontrivial Fourier terms have period 6 in j , and the closed form naturally splits into six residue classes,

$$(3.8) \quad 3S_{n,j} = \begin{cases} \delta(2^j - 3j - 1) - 3j(j-1), & j \equiv 0 \pmod{6}, \\ \delta(2^j - 2) - 3j(j-1), & j \equiv 1 \pmod{6}, \\ \delta(2^j + 3j - 1), & j \equiv 2 \pmod{6}, \\ \delta(2^j + 3j + 1) + 3j(j-1), & j \equiv 3 \pmod{6}, \\ \delta(2^j + 2) + 3j(j-1), & j \equiv 4 \pmod{6}, \\ \delta(2^j - 3j + 1), & j \equiv 5 \pmod{6}. \end{cases}$$

In detail, for

$$R_a(j) = \sum_{\nu \equiv a \pmod{3}} \binom{j}{\nu},$$

the root-of-unity filter gives

$$R_a(j) = \frac{1}{3} \sum_{\lambda=0}^2 \omega^{-a\lambda} (1 + \omega^\lambda)^j = \frac{1}{3} \left(2^j + 2 \cos \frac{(j-2a)\pi}{3} \right),$$

where $2 \cos((j-2a)\pi/3)$ equals $2, 1, -1, -2, -1, 1$ according as $j-2a \equiv 0, 1, 2, 3, 4, 5 \pmod{6}$. The first two moments over the residue class are

$$\sum_{\nu \equiv 2 \pmod{3}} \nu \binom{j}{\nu} = j R_1(j-1), \quad \sum_{\nu \equiv 2 \pmod{3}} \nu^2 \binom{j}{\nu} = j(j-1) R_0(j-2) + j R_1(j-1).$$

Expanding the summand in (3.7) gives, for $j \geq 2$,

$$(3.9) \quad \begin{aligned} S_{n,j} &= \delta(2j R_1(j-1) + (1-j) R_2(j)) \\ &\quad + j(j-1)(-2R_0(j-2) + 3R_1(j-1) - R_2(j)), \end{aligned}$$

with $j = 0, 1$ following directly from (3.7). Substituting the values of $R_2(j)$, $R_1(j - 1)$, and $R_0(j - 2)$ by residue $j \bmod 6$ yields (3.8).

It remains only to inspect the six cases. The cases $j \equiv 2, 3, 4 \pmod{6}$ are immediate. If $j \equiv 5 \pmod{6}$, then $j \geq 5$ and $2^j - 3j + 1 > 0$. For the two remaining classes, note that $n + 1 = 3m - 1$ is congruent to 2 or 5 modulo 6; hence $\delta \geq 1$ when $j \equiv 1 \pmod{6}$, and $\delta \geq 2$ when $j \equiv 0 \pmod{6}$ and $j > 0$. If $j \equiv 1 \pmod{6}$, the case $j = 1$ gives zero, while for $j \geq 7$,

$$(3.10) \quad 2^j - 2 \geq 3j(j - 1).$$

Finally, when $j \equiv 0 \pmod{6}$ and $j > 0$, one has $j \geq 6$ and

$$(3.11) \quad 2(2^j - 3j - 1) \geq 3j(j - 1).$$

The inequalities (3.10) and (3.11) hold with equality at $j = 7$ and $j = 6$, respectively. Advancing j by one increases the left-minus-right difference by $2^j - 6j$ in (3.10) and by $2^{j+1} - 6j - 6$ in (3.11), both positive for $j \geq 6$; the inequalities therefore persist for all larger j . Likewise $2^j - 3j + 1 > 0$ at $j = 5$, with increment $2^j - 3 > 0$ thereafter. Hence $S_{n,j} \geq 0$ in every case. Therefore $b_j \geq 0$ for every j . At $j = 2$, since $\delta = n - 1$, one has $3S_{n,2} = 9(n - 1)$, and hence

$$b_2 = \frac{n - 1}{n + 1} > 0.$$

Equivalently, all coefficients of $\mathcal{P}_m$ are nonnegative and its coefficient at ξ^2 is positive. Therefore $\mathcal{P}_m(\xi) > 0$ for $\xi > 0$. The substitution $t = \xi/(1 + \xi)$ in (3.4) proves $J_m(t) > 0$ for $0 < t < 1$, and (3.2) completes the proof. $\square$

3.2. Insertion of monotone weights. The constant-weight certificate is enough because the admissible weight cone is generated by symmetric central windows. The following summation-by-parts identity characterizes the relevant dual cone. Its one-sign-change form is the standard weighting device used in this circle of coefficientwise Turán inequalities; compare [3, Lem. 6] and [2, Lem. 2.1].

Lemma 3.3. *Let $A_1, \dots, A_L \in \mathbb{R}$, put*

$$T_\sigma = \sum_{k=\sigma}^L A_k \quad (1 \leq \sigma \leq L),$$

and let $0 \leq w_1 \leq \dots \leq w_L$, with $w_0 = 0$. Then

$$(3.12) \quad \sum_{k=1}^L w_k A_k = \sum_{\sigma=1}^L (w_\sigma - w_{\sigma-1}) T_\sigma.$$

Consequently,

$$\sum_{k=1}^L w_k A_k \geq 0 \quad \text{for every } 0 \leq w_1 \leq \dots \leq w_L$$

if and only if $T_\sigma \geq 0$ for every σ . If all $T_\sigma > 0$, the weighted sum is strictly positive for every nonzero such weight sequence.

In particular, if $A_1, \dots, A_L$ has no more than one sign change, from nonpositive to nonnegative, and $\sum_k A_k \geq 0$, then all $T_\sigma \geq 0$. If moreover $\sum_k A_k > 0$ and $A_L > 0$, then all $T_\sigma > 0$.

Proof. Since $A_k = T_k - T_{k+1}$, with $T_{L+1} = 0$, summation by parts gives (3.12). The cone equivalence follows because the increments $w_\sigma - w_{\sigma-1}$ are nonnegative, while choosing $w_k = \mathbf{1}_{\{k \geq \sigma\}}$ recovers the individual tail T_σ .

For the final assertion, suppose the signs change at most once from nonpositive to nonnegative. After the crossing, each T_σ is a sum of nonnegative terms. Before the crossing,

$$T_\sigma = \sum_{k=1}^L A_k - \sum_{k < \sigma} A_k \geq \sum_{k=1}^L A_k \geq 0,$$

because the omitted prefix is nonpositive. If the total sum is positive and $A_L > 0$, the same argument is strict before the crossing, while every tail after it contains a positive term. Hence all $T_\sigma > 0$. $\square$

After pairing complementary residue blocks, a radial one-sign-change property shows that every central window has positive derivative. The duality lemma then inserts arbitrary weights increasing toward the center.

For $0 < t < 1$, differentiation of (2.5) gives

$$(3.13) \quad \frac{d}{dt} G_{m,w} \left(\frac{t}{1+t} \right) = \frac{3J_{m,w}(t)}{(1+t)^{3m+1}},$$

where

$$(3.14) \quad J_{m,w}(t) = \sum_{k=1}^{m-1} w_k \binom{3m-2}{3k-1} t^{3k-1} (k - (m-k)t).$$

Pair the terms indexed by k and $m-k$, as in the proof of [6, Thm. 2]. For $1 \leq k < m/2$, let

$$(3.15) \quad B_{m,k}(t) = \binom{3m-2}{3k-1} \left[t^{3k-1} (k - (m-k)t) + t^{3(m-k)-1} ((m-k) - kt) \right].$$

If m is even, the central block is

$$(3.16) \quad B_{m,m/2}(t) = \binom{3m-2}{3m/2-1} \frac{m}{2} t^{3m/2-1} (1-t) > 0.$$

Then

$$J_{m,w}(t) = \sum_{k=1}^{\lfloor m/2 \rfloor} w_k B_{m,k}(t),$$

with the convention (3.16) at the center.

Lemma 3.4. *For each fixed $m \geq 2$ and $0 < t < 1$, the sequence*

$$B_{m,1}(t), B_{m,2}(t), \dots, B_{m,\lfloor m/2 \rfloor}(t)$$

has no more than one sign change, from negative to positive, and its final term $B_{m,\lfloor m/2 \rfloor}(t)$ is positive.

Proof. For $k < m/2$, the sign of $B_{m,k}(t)$ is the sign of

$$H_{m,k}(t) = k - (m-k)t + t^{3(m-2k)}((m-k) - kt).$$

The natural radial coordinate of the paired block is its distance $\eta = m - 2k > 0$ from the center. Put $t = e^{-z}$, $z > 0$. Direct rearrangement gives

$$(3.17) \quad \begin{aligned} H_{m,k}(t) &= \frac{1}{2}(1+t)(1+t^{3\eta}) \left[m \frac{1-t}{1+t} - \eta \frac{1-t^{3\eta}}{1+t^{3\eta}} \right] \\ &= \frac{1}{2}(1+t)(1+t^{3\eta}) \left[m \tanh \frac{z}{2} - \eta \tanh \frac{3\eta z}{2} \right]. \end{aligned}$$

The function $\eta \mapsto \eta \tanh(3\eta z/2)$ is strictly increasing for $\eta > 0$. Thus moving outward from the center, which increases η and decreases k , can only decrease the bracket in (3.17). In particular, for $2 \leq k < m/2$, if $B_{m,k}(t) \leq 0$, then the bracket is strictly negative after replacing η by $\eta + 2$, equivalently k by $k - 1$. Hence

$$B_{m,k}(t) \leq 0 \implies B_{m,k-1}(t) < 0 \quad (2 \leq k < m/2).$$

The final block is always positive. For even m this is (3.16). For odd m , the final block has $\eta = 1$, and its bracket in (3.17) is $m \tanh(z/2) - \tanh(3z/2) > 0$, because $m \geq 3$ and $\tanh(3y) < 3 \tanh y$ for $y > 0$. Hence the blocks have the asserted sign pattern. $\square$

Proof of Theorem 3.1. Put $L = \lfloor m/2 \rfloor$ and define the central-window tails

$$T_{m,\sigma}(t) = \sum_{k=\sigma}^L B_{m,k}(t), \quad 1 \leq \sigma \leq L.$$

By Lemma 3.2,

$$T_{m,1}(t) = J_m(t) > 0 \quad (0 < t < 1).$$

By Lemma 3.4, the block sequence has at most one sign change from negative to positive and $B_{m,L}(t) > 0$. The strict one-sign-change consequence of Lemma 3.3 therefore gives

$$T_{m,\sigma}(t) > 0 \quad (1 \leq \sigma \leq L, 0 < t < 1).$$

These tails are precisely the J -numerators corresponding to the extreme central-window weights $\mathbf{1}_{\{k \geq \sigma\}}$ in the paired indexing.

Now set $w_0 = 0$. The duality identity (3.12) gives

$$J_{m,w}(t) = \sum_{\sigma=1}^L (w_\sigma - w_{\sigma-1}) T_{m,\sigma}(t).$$

Every increment is nonnegative by (3.1); hence $J_{m,w}(t) \geq 0$, and it is strictly positive for $0 < t < 1$ whenever the weights are not all zero. Equation (3.13) proves the asserted nonstrict and strict monotonicity of $G_{m,w}$ on $[0, 1/2]$. $\square$

The step weight $\mathbf{1}_{\{k \geq \sigma\}}$ retains exactly the central index window $\sigma \leq k \leq m - \sigma$; the cone of centrally nondecreasing symmetric weights is the positive span of these windows, with coefficients $w_\sigma - w_{\sigma-1}$. Thus the constant-weight certificate and the radial one-sign-change property first prove positivity for every central truncation, and monotone weighting is then only a cone decomposition.

4. PROOF OF THE CUBIC THEOREM

With the exact reduction and the cubic kernel theorem in place, the main result is an immediate degreewise consequence.

Corollary 4.1. *Fix $m \geq 2$, and let (w_k) satisfy (3.1). Then $(u, v) \mapsto C_{m,w}(u, v)$ is Schur-concave on $[0, \infty)^2$. Equivalently, at fixed $s > 0$,*

$$d \mapsto C_{m,w} \left(\frac{s}{2} + d, \frac{s}{2} - d \right)$$

is nonincreasing for $0 \leq d \leq s/2$, and is strictly decreasing on this interval if the weights are not all zero.

Proof. Apply Theorem 3.1 and Proposition 2.4, including their strict parts. $\square$

Proof of Theorem 1.1. The zero sequence is trivial, so assume $f \not\equiv 0$. The coefficients of degrees 0 and 1 vanish. Fix $m \geq 2$ and set

$$(4.1) \quad w_k = f_k f_{m-k}, \quad 1 \leq k \leq m-1.$$

These weights are symmetric, and Lemma 2.1 gives (3.1). Direct multiplication gives

$$(4.2) \quad \begin{aligned} [x^m] \mathcal{S}_f(s; d_1, d_2; x) &= C_{m,w} \left(\frac{s}{2} + d_1, \frac{s}{2} - d_1 \right) \\ &\quad - C_{m,w} \left(\frac{s}{2} + d_2, \frac{s}{2} - d_2 \right). \end{aligned}$$

This is nonnegative by Corollary 4.1, which proves the fixed-sum statement. The equivalent Turánian form follows from the parameter correspondence after Theorem 1.1; the case $s = 0$ is trivial. $\square$

The strict part of the kernel theorem also determines exactly where the coefficient inequalities are nontrivial.

Corollary 4.2. *Under the hypotheses of Theorem 1.1, put $I = \{n \geq 1 : f_n > 0\}$ and $I + I = \{i + j : i, j \in I\}$. If $s > 0$ and $0 \leq d_1 < d_2 \leq s/2$, then*

$$[x^m]\mathcal{S}_f(s; d_1, d_2; x) > 0 \iff m \in I + I \quad (m \geq 2).$$

Thus if $I = \{a, \dots, b\}$ the positive degrees are exactly $2a, \dots, 2b$, while if $I = \{a, a + 1, \dots\}$ they are exactly $m \geq 2a$. The same description holds for (1.4) whenever $\alpha, \beta > 0$.

Proof. The condition $m \in I + I$ is exactly that the weights (4.1) are not all zero. If they all vanish, so does (4.2); otherwise strictness follows directly from Corollary 4.1. Since I is an integer interval, so is $I + I$. The Turánian statement follows from the parameter equivalence and the fact that $\alpha, \beta > 0$ exactly corresponds to $d_1 < d_2$. $\square$

Remark 4.3. At a fixed degree m , only the single chain

$$f_1 f_{m-1} \leq f_2 f_{m-2} \leq \dots \leq f_{\lfloor m/2 \rfloor} f_{\lceil m/2 \rceil}$$

is needed. Thus the degree- m conclusions hold for any nonnegative sequence satisfying that local condition. Globally, however, Lemma 2.1 shows that log-concavity together with interval support is not merely a convenient sufficient hypothesis: it is exactly the condition that all of these anti-diagonal product chains are centrally nondecreasing simultaneously. Interval support also makes $I + I$ an interval in Corollary 4.2.

5. THE SHARP MULTIPLICITY THRESHOLD

The cubic theorem closes the only unresolved universal case. Degree three already explains why no larger multiplicity can satisfy the same statement.

Proposition 5.1 (Degree-three multiplicity threshold). *For every integer $r \geq 2$, the degree-three constant-weight kernel is*

$$G_3^{(r)}(p) = \binom{3r-2}{r-1} [p(1-p)]^r (p^r + (1-p)^r),$$

and, writing it as $\widehat{G}_3^{(r)}(q)$ with $q = p(1-p)$,

$$(5.1) \quad \left. \frac{d}{dq} \widehat{G}_3^{(r)}(q) \right|_{q=1/4} = \binom{3r-2}{r-1} r(3-r) 2^{2-3r}.$$

For the two-term sequence $f_1 = f_2 = 1$, $f_n = 0$ for $n \geq 3$,

$$(5.2) \quad \begin{aligned} & [x^3] \left(F_{f,r}(\mu+1; x)^2 - F_{f,r}(\mu; x) F_{f,r}(\mu+2; x) \right) \\ &= \frac{r(4r - (r-3)\mu)(\mu+2)_{r-2}(\mu+1)_{2r-1}}{(r-1)!(2r-1)!}. \end{aligned}$$

Hence for every $r \geq 4$ this coefficient is negative exactly when $\mu > 4r/(r-3)$. Moreover, this counterexample is minimal both in degree and in positive-support size: degree two is nonnegative for every r , and no one-point-support sequence can produce a negative coefficient.

Dividing each kernel by its central value puts the multiplicities $r = 2, \dots, 5$ on one scale (Figure 1).

Proof. For general multiplicity r , the reduction (2.2)–(2.5) runs with 3 replaced by r throughout, giving the constant-weight kernel

$$G_m^{(r)}(p) = p(1-p) \mathbb{P}\{\text{Bin}(rm-2, p) \equiv r-1 \pmod{r}\}.$$

At the first degree with a nontrivial complementary pair, $m = 3$, only the residues corresponding to $k = 1, 2$ occur, which gives the closed form for $G_3^{(r)}$ asserted in the proposition. Put $p = 1/2 + x$ and $q = p(1-p) = 1/4 - x^2$. Expansion at the center gives

$$G_3^{(r)}\left(\frac{1}{2} + x\right) = \binom{3r-2}{r-1} (2^{1-3r} + r(r-3)2^{2-3r}x^2 + O(x^4)),$$

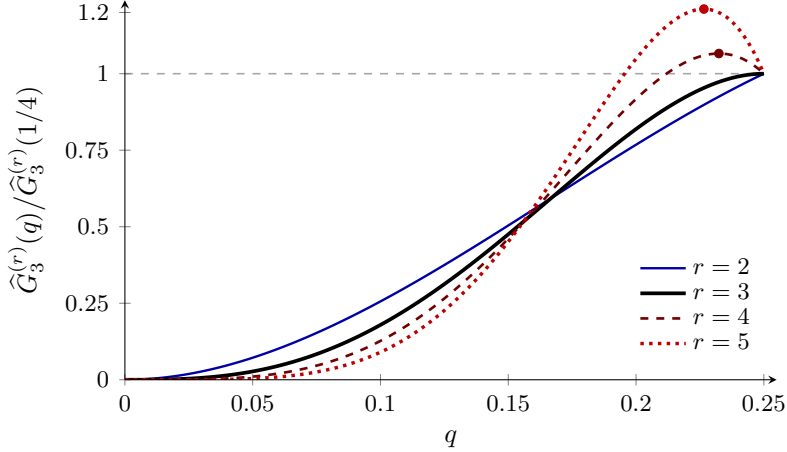


FIGURE 1. The degree-three constant-weight kernels of Proposition 5.1, each divided by its central value $\widehat{G}_3^{(r)}(1/4) = \binom{3r-2}{r-1} 2^{1-3r}$, on $0 \leq q \leq 1/4$. Under this normalization (5.1) becomes the endpoint slope $2r(3-r)$, so the kernel rises to its central maximum at $r = 2$, arrives tangentially at $r = 3$, and for $r \geq 4$ overshoots and returns downward to the center. Since $1/4 - q = (p - 1/2)^2$, the tangency at $r = 3$ is the quartically flat maximum at $p = 1/2$. The marked interior maxima at $r = 4, 5$ exceed the central values by 6.59% and 21.12%.

which is equivalent to (5.1). Thus the center is a quadratic maximum for $r = 2$, while for every $r \geq 4$ it is a quadratic local minimum. At the critical value $r = 3$ the quadratic term cancels, but

$$G_3^{(3)}(p) = 21q^3(1 - 3q), \quad \frac{d}{dq}G_3^{(3)} = 63q^2(1 - 4q) > 0 \quad (0 < q < 1/4),$$

and

$$G_3^{(3)}\left(\frac{1}{2} + x\right) = \frac{21}{256} - \frac{63}{8}x^4 + O(x^6).$$

Hence the degree-three cubic center is a quartically flat maximum: the first local obstruction is exactly neutral at quadratic order, and Theorem 3.1 supplies the global positivity that this local calculation alone cannot see.

For the stated two-term sequence, direct multiplication gives

$$\begin{aligned} & [x^3] \left(F_{f,r}(\mu + 1; x)^2 - F_{f,r}(\mu; x) F_{f,r}(\mu + 2; x) \right) \\ &= \frac{2(\mu + 1)_r(\mu + 1)_{2r} - (\mu)_r(\mu + 2)_{2r} - (\mu)_{2r}(\mu + 2)_r}{(r - 1)!(2r - 1)!}, \end{aligned}$$

Factoring out $(\mu + 2)_{r-2}(\mu + 1)_{2r-1}$ leaves $r(4r - (r - 3)\mu)$, giving (5.2). The extracted Pochhammer factors are positive for $\mu \geq 0$, so for $r \geq 4$ the sign changes exactly at $\mu = 4r/(r - 3)$. The repeated factor $r - 3$ in (5.1) and (5.2) reflects the same mechanism: as μ grows, the beta laws in the fixed-sum representation concentrate near $q = 1/4$, so the finite-shift comparison increasingly probes the local central behavior of the kernel. At $r = 4$ the threshold is $\mu = 16$, giving an exact counterpart to the numerical observation of [6, Rem. 5].

Finally, for every integer $N \geq 1$,

$$\frac{d^2}{d\mu^2} \log(\mu)_N = - \sum_{j=0}^{N-1} \frac{1}{(\mu + j)^2} < 0 \quad (\mu > 0),$$

so $\mu \mapsto (\mu)_N$ is log-concave and its generalized Turánian is nonnegative (with the boundary $\mu = 0$ immediate). The degree-two coefficient is a nonnegative multiple of this Turánian with $N = r$. If f has one-point support $\{a\}$, its only active product degree is $2a$, where the same

argument applies with $N = ra$. Thus neither a lower degree nor a smaller positive support can furnish a counterexample. $\square$

Corollary 5.2 (Sharp multiplicity classification). *For integers $r \geq 2$, the family $F_{f,r}$ in (1.1) is coefficientwise log-concave for every nonnegative log-concave sequence with no internal zeros if and only if*

$$r \in \{2, 3\}.$$

Proof. The case $r = 2$ is [6, Thm. 2], and the case $r = 3$ is Theorem 1.1. For every $r \geq 4$, the two-term sequence in Proposition 5.1 satisfies all sequence hypotheses, while (5.2) is negative for $\mu > 4r/(r - 3)$. $\square$

5.1. The first supercritical case. The Bernstein-basis positivity certificate of Lemma 3.2 can fail even when the kernel remains monotone. In the analogue $(1 + \xi)^{4m-1} J_m^{(4)}(\xi/(1 + \xi))$, the coefficients at ξ^7 and ξ^8 are

$$\begin{aligned} a_7 &= \frac{8}{315}(m-2)(m-1)(2m-3)(2m-1)(4m-7)(4m-5)(4m-3), \\ a_8 &= -\frac{7}{2}a_7, \end{aligned}$$

and $a_7 > 0$ precisely for $m \geq 3$, so the certificate fails there. Nevertheless, at $m = 4$, with $z = (1 - t)^2/t > 0$,

$$J_4^{(4)}(t) = 52t^7(1 - t)(7z^4 + 42z^3 + 42(z - 1)^2 + 6) > 0 \quad (0 < t < 1).$$

Thus failure of this Bernstein certificate is not equivalent to failure of kernel monotonicity. The first supercritical multiplicity therefore retains positive structure beyond the universal theorem.

6. CONSEQUENCES

6.1. Hypergeometric specialization. For $f_n \equiv 1$, the triplication formula gives

$$F_f(\mu; x) = 3x \frac{d}{dx} {}_3F_2\left(\frac{\mu}{3}, \frac{\mu+1}{3}, \frac{\mu+2}{3}; \frac{1}{3}, \frac{2}{3}; x\right).$$

Hence Theorem 1.1 yields coefficientwise generalized Turán inequalities for this differentiated ${}_3F_2$ family, while general log-concave (f_n) gives a coefficient-weighted extension. Related coefficientwise inequalities for generalized hypergeometric functions under simultaneous parameter shifts are studied in [4, Thm. 3].

6.2. Strict pointwise log-concavity. The exact support of Corollary 4.2 upgrades the pointwise Turán inequality from nonnegativity to strictness.

Corollary 6.1. *Suppose (f_n) is not identically zero, and let R be the radius of convergence of $\sum_{n \geq 1} f_n x^n$. For every $0 < x < R$, the function $\mu \mapsto F_f(\mu; x)$ is finite, continuous, and positive on $(0, \infty)$, and $\mu \mapsto \log F_f(\mu; x)$ is strictly concave there. More generally, for $\mu \geq 0$ and $\alpha, \beta > 0$,*

$$F_f(\mu + \alpha; x)F_f(\mu + \beta; x) > F_f(\mu; x)F_f(\mu + \alpha + \beta; x).$$

Proof. Log-concavity with no internal zeros makes f_{n+1}/f_n nonincreasing on the positive support. If that support is finite, then $R = \infty$. Otherwise it has the form $\{a, a + 1, \dots\}$, and

$$f_n \leq f_a \left(\frac{f_{a+1}}{f_a} \right)^{n-a} \quad (n \geq a),$$

so again $R > 0$. For fixed $\mu > 0$,

$$\frac{(\mu)_{3n}}{(3n-1)!} = \frac{\Gamma(3n + \mu)}{\Gamma(\mu)\Gamma(3n)} \sim \frac{(3n)^\mu}{\Gamma(\mu)},$$

so $F_f(\mu; \cdot)$ has radius R .

For local uniform convergence in μ , fix $M > 0$ and $0 < x < R$. For $0 \leq \mu \leq M$,

$$\frac{(\mu)_{3n}}{(3n-1)!} = \mu \prod_{j=1}^{3n-1} \left(1 + \frac{\mu}{j} \right) \leq M \exp \left(M \sum_{j=1}^{3n-1} \frac{1}{j} \right) \leq M e^M (3n)^M.$$

Choose y with $x < y < R$. Since $n^M(x/y)^n$ is bounded, the summands of $F_f(\mu; x)$ are uniformly dominated for $0 \leq \mu \leq M$ by a constant multiple of $f_n y^n$. The Weierstrass test gives local uniform convergence, hence continuity; positivity is immediate from the coefficients.

Choose n_0 with $f_{n_0} > 0$. By Corollary 4.2, the coefficient of x^{2n_0} in the generalized Turánian is positive for $\alpha, \beta > 0$, while all coefficients are nonnegative by Theorem 1.1. Summing gives the strict inequality. Taking $\alpha = \beta = h > 0$ yields strict midpoint concavity of $\log F_f(\mu; x)$; continuity gives concavity, and equality at any interior convex combination would force affinity on an interval, contradicting strict midpoint concavity. $\square$

6.3. Differential Turán inequality.

Corollary 6.2. *Under the hypotheses of Theorem 1.1, for every $\mu \geq 0$ and every $m \geq 2$,*

$$[x^m] \left[(\partial_\mu F_f(\mu; x))^2 - F_f(\mu; x) \partial_\mu^2 F_f(\mu; x) \right] > 0 \iff m \in I + I,$$

where $I = \{n \geq 1 : f_n > 0\}$. In particular the differential Turánian has nonnegative coefficients, with the same exact positive support as the finite-shift Turánians of Corollary 4.2.

Proof. Fix $m \geq 2$ and use the weights $w_k = f_k f_{m-k}$ from (4.1). If $m \notin I + I$, all these weights vanish, so the degree- m coefficient is zero. Assume $m \in I + I$. By Lemma 2.1, these weights satisfy (3.1), and they are not all zero.

First let $\mu > 0$. For these weights $C_{m,w}(u, v) = [x^m] F_f(u; x) F_f(v; x)$ by Remark 2.5, so put

$$\Phi_m(d) = C_{m,w}(\mu + d, \mu - d), \quad |d| < \mu.$$

Differentiating the symmetric product at $d = 0$ gives

$$[x^m] \left[(\partial_\mu F_f)^2 - F_f \partial_\mu^2 F_f \right] = -\frac{1}{2} \Phi_m''(0).$$

The fixed-sum beta representation (2.6) remains valid for $|d| < \mu$, so

$$\Phi_m(d) = \kappa_{m,2\mu} \mathbb{E} \widehat{G}_{m,w}(Q_d).$$

Use the density g_d from the proof of Lemma 2.3. Its formula is even in d and has the form

$$g_d(q) = c(d) h(q) \cosh(dl(q)),$$

with $c(d)$ independent of q . Fix $0 < d_0 < \mu$. Uniformly for $|d| \leq d_0$, g_d and its first two d -derivatives are $O(q^{\mu-1-d_0} \log^2(1/q))$ as $q \downarrow 0$ and $O((1-4q)^{-1/2})$ as $q \uparrow 1/4$, since $\ell(q) = \log(1/q) + O(q)$ at the first endpoint and $\ell(q) = O(\sqrt{1-4q})$ at the second. Both bounds are integrable, so we may differentiate twice under the integral at $d = 0$. Using $\int g_d = 1$ gives

$$\partial_d^2 g_d(q) \big|_{d=0} = g_0(q) (\ell(q)^2 - \mathbb{E} \ell(Q_0)^2).$$

Consequently

$$\Phi_m''(0) = \kappa_{m,2\mu} \text{Cov} \left(\widehat{G}_{m,w}(Q_0), \ell(Q_0)^2 \right).$$

Hence Theorem 3.1 makes $\widehat{G}_{m,w}$ strictly increasing on $(0, 1/4)$, whereas $\ell(q)^2$ is strictly decreasing there. If Q'_0 is an independent copy of Q_0 , then

$$\begin{aligned} & 2 \text{Cov}(\widehat{G}_{m,w}(Q_0), \ell(Q_0)^2) \\ &= \mathbb{E}[(\widehat{G}_{m,w}(Q_0) - \widehat{G}_{m,w}(Q'_0))(\ell(Q_0)^2 - \ell(Q'_0)^2)]. \end{aligned}$$

The product is nonpositive and, since Q_0 is nondegenerate, strictly negative with positive probability. Hence $\Phi_m''(0) < 0$, proving strict positivity.

Finally, at $\mu = 0$,

$$F_f(0; x) = 0, \quad \partial_\mu F_f(0; x) = \sum_{n \geq 1} f_n x^n,$$

because

$$\partial_\mu \frac{(\mu)_{3n}}{(3n-1)!} \bigg|_{\mu=0} = 1.$$

Thus the differential Turánian is $(\sum_{n \geq 1} f_n x^n)^2$, whose positive coefficient support is exactly $I + I$. $\square$

7. CONCLUDING REMARKS

The cubic case is the endpoint of universal coefficientwise log-concavity for the multiple-Pochhammer family. Log-concavity of the input sequence is exactly central increase of its anti-diagonal products; the beta mixture turns balance of the parameters into stochastic centrality; and the cubic residue kernel preserves that order. These three levels fit together without loss: beta concentration shows that the kernel is the exact fixed-degree obstruction. At degree three the central slope changes sign through the factor $3 - r$, so the same mechanism also explains the sharp multiplicity threshold.

Beyond that threshold, universal positivity fails, but failure of the Bernstein certificate does not force failure of kernel monotonicity. Section 5 shows this already for $r = 4$. Two natural questions remain: does the constant-weight kernel at multiplicity four stay monotone in every degree, and which restricted cones of centrally weighted sequences continue to preserve monotonicity for multiplicities beyond three? These questions separate the failure of a universal theorem from the finer positive structure that may survive in the supercritical regime.

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Data availability. No datasets were generated or analyzed.

Computational verification. The proofs do not depend on numerical computation. Scripts used for verification are available at <https://github.com/johnnyshields/papers>.

Generative AI use. During the preparation of this work, the author used OpenAI's ChatGPT (GPT-5.6 Sol) and Anthropic's Claude (Opus 5, Opus 4.8) to assist with mathematical research, manuscript drafting, and the development of auxiliary verification code. After using these tools, the author reviewed and edited as needed. The author takes full responsibility for the content of the work.

REFERENCES

- [1] K. S. Berenhaut and L. D. Bergen, *Stochastic orderings, folded beta distributions and fairness in coin flips*, Statist. Probab. Lett. **81** (2011), no. 6, 632–638, DOI: 10.1016/j.spl.2011.01.002.
- [2] S. I. Kalmykov and D. B. Karp, *Log-concavity for series in reciprocal gamma functions and applications*, Integral Transforms Spec. Funct. **24** (2013), no. 11, 859–872, DOI: 10.1080/10652469.2013.764874.
- [3] S. I. Kalmykov and D. B. Karp, *Log-convexity and log-concavity for series in gamma ratios and applications*, J. Math. Anal. Appl. **406** (2013), no. 2, 400–418, DOI: 10.1016/j.jmaa.2013.04.061.
- [4] S. I. Kalmykov and D. B. Karp, *Log-concavity and Turán-type inequalities for the generalized hypergeometric function*, Anal. Math. **43** (2017), no. 4, 567–580, DOI: 10.1007/s10476-017-0503-z.
- [5] D. Karp and S. M. Sitnik, *Log-convexity and log-concavity of hypergeometric-like functions*, J. Math. Anal. Appl. **364** (2010), no. 2, 384–394, DOI: 10.1016/j.jmaa.2009.10.057.
- [6] D. Karp and Y. Zhang, *Log-concavity and log-convexity of series containing multiple Pochhammer symbols*, Fract. Calc. Appl. Anal. **27** (2024), no. 1, 458–486, DOI: 10.1007/s13540-023-00238-0.
- [7] V. E. Levit and E. Mandrescu, *Some operations preserving log-concavity of nonnegative functions*, Math. Inequal. Appl. **9** (2006), no. 1, 63–73, DOI: 10.7153/mia-09-07.
- [8] A. W. Marshall, I. Olkin, and B. C. Arnold, *Inequalities: Theory of Majorization and Its Applications*, 2nd ed., Springer Series in Statistics, Springer, New York, 2011, DOI: 10.1007/978-0-387-68276-1.
- [9] A. Ninh and M. Pham, *Logconcavity, twice-logconcavity and Turán-type inequalities*, Ann. Oper. Res. **354** (2025), no. 3, 1271–1283, DOI: 10.1007/s10479-018-2923-y.
- [10] V. Powers and B. Reznick, *Polynomials that are positive on an interval*, Trans. Amer. Math. Soc. **352** (2000), no. 10, 4677–4692, DOI: 10.1090/S0002-9947-00-02595-2.
- [11] M. Shaked and J. G. Shanthikumar, *Stochastic Orders*, Springer Series in Statistics, Springer, New York, 2007, DOI: 10.1007/978-0-387-34675-5.